

One Peg, Two Shocks: Monetary and Real Transmission – BVAR Evidence in Hong Kong

Steven Chung

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Abstract

Under Hong Kong’s Linked Exchange Rate System, U.S. monetary conditions are transmitted directly into domestic funding costs, while Mainland China provides a separate external real-demand channel. This paper estimates a Bayesian VAR with exogenous U.S. and China variables to study how these shocks transmit through Hong Kong’s economy. Two results stand out. First, monetary transmission is asymmetric in speed: higher HIBOR depresses property prices quickly, but the effect on output persists longer through credit and debt-servicing conditions. Second, China-related external demand supports Hong Kong real output, but within the domestic system the property channel is the larger amplifier of GDP fluctuations under the baseline recursive identification. Overall, the evidence suggests that Hong Kong absorbs imported U.S. financial conditions rapidly under the peg, and that domestic property and credit markets shape how those shocks feed into real activity.

1 Introduction

The transmission of external shocks in small open economies is a classic topic in international macroeconomics. With a fixed exchange rate, foreign monetary policy shocks can directly affect the domestic economy through various channels. Under Hong Kong’s Linked Exchange Rate System (LERS), changes in the U.S. federal funds rate are transmitted to Hong Kong through the HKMA Base Rate and HKD liquidity condition, with the three-month HIBOR serving as the primary market rate through which this impulse reaches the domestic economy.

This paper asks: through which domestic channels do U.S. interest rate movements transmit to Hong Kong macroeconomic variables, especially real GDP, and with what speed and persistence? Beyond U.S. monetary shocks, how do other external shocks, such as Mainland China’s real GDP fluctuations, transmit to Hong Kong macroeconomic variables, and through which channels?

This paper explores these questions with a Bayesian Vector Autoregressive model with exogenous variables (BVARX) of order 4 estimated on quarterly data from 1998 Q1 to 2026 Q1. The model incorporates six endogenous variables, which are the three-month HIBOR, exports to China, property prices, GDP growth, CPI inflation, and the unemployment rate. Following the block-exogenous structure of Cushman and Zha (1997), the US federal funds rate and China’s GDP growth are included as (strictly) exogenous variables. The analysis relies on impulse response functions (IRFs) and forecast error variance decompositions (FEVDs) to quantify the transmission of external and internal shocks.

The main results are as follows. First, the estimated contemporaneous coefficient on `us_ffr` in the HIBOR equation is positive and significant, consistent with the LERS pass-through idea; HIBOR’s 94% own-share in its FEVD reflects that other domestic shocks account for little of its variation. Second, a HIBOR tightening depresses property prices significantly within one quarter but the response crosses zero by $h = 2$. Third, despite this short-lived property response, property price shocks are the dominant source of GDP variance at the eight-quarter horizon (20.7%), larger than the direct trade channel (exports: 16.4%) or direct monetary channel (HIBOR: 8.4%). Fourth, the direct HIBOR–GDP link is significant at horizons one through four, indicating a credit and mortgage sub-channel that sustains output contraction beyond the asset price effect. Fifth, exports to China explain 16.4% of GDP forecast error variance in the BVAR and are significant at $h = 1$ and $h = 2$; in recursive LP, this channel weakens after conditioning on current China GDP growth. This suggests that China’s real GDP affects Hong Kong output through the export channel.

The paper makes two contributions. First, it documents a speed asymmetry within monetary transmission under a currency board: the property sub-channel is fast (significant at $h = 1$ – 2) but short-lived, while the direct credit sub-channel is slower but more persistent (significant through $h = 4$). Mishkin (1996) notes that interest rate and asset price channels operate on different timescales; our FEVD and IRF evidence quantifies this asymmetry under the institutional conditions of LERS. Second, the paper shows that domestic property amplification accounts for a larger share of GDP forecast error variance (20.7%) than either the monetary channel (HIBOR) (8.4%) or the trade channel (16.4%) at an eight-quarter horizon, under recursive identification (orthogonalized shocks) with HIBOR ordered first.

The remainder of the paper is organized as follows. Section 2 discusses the institutional background of Hong Kong. Section 3 describes the data and transformation. Section 4 presents the empirical methodology. Section 5 presents the main results. Section 6 reports robustness checks and some extensions. Section 7 documents some limitations and future steps.

2 Institutional Background and Channels

Currency board

Hong Kong's Linked Exchange Rate System (LERS) has operated as a currency board since 1983, and the Hong Kong dollar (HKD) was pegged at HKD 7.80 per US dollar. Under a 2005 refinement, the HKMA maintains two convertibility undertakings: it will sell US dollars at 7.75 (strong-side) and buy at 7.85 (weak-side), anchoring the exchange rate in a narrow band around the 7.80 central parity.

The Base Rate mechanism

Theoretically, with a fixed exchange rate, when the Federal Reserve adjusts the U.S. federal funds rate (FFR), USD rates go up globally. Investors sell HKD and buy USD to earn higher returns, putting downward pressure on the Hong Kong dollar. The HKMA defends the peg by selling USD and buying HKD to support the exchange rate. That drains HKD from Hong Kong's banking system, which leads to a higher funding cost domestically. Banks then have less HKD to lend each other, leading to higher HIBOR across maturities.

Technically, the HKMA sets the Base Rate mechanically as the maximum of FFR's lower bound plus 50 bps and the 5-day moving average of overnight & 1M HIBOR. When the Federal Reserve moves its target, the HKMA Base Rate adjusts automatically without any HKMA decision. The Base Rate governs the terms at which the HKMA lends to licensed banks through its discount window, anchoring the upper limit of overnight interbank rate.

This paper uses the three-month HIBOR rather than the Base Rate because HIBOR reflects actual market borrowing costs at the three-month horizon. It is market-determined but tracks US monetary conditions closely, as the liquidity contraction mechanism described above leaves banks with little room to deviate from the imported rate cycle.

Monetary policy autonomy under the trilemma.

LERS, combined with Hong Kong's open capital account, eliminates HK's policy autonomy through the Mundell-Fleming trilemma. With a fixed exchange rate and free capital mobility, Hong Kong cannot independently set domestic interest rates. When the Federal Reserve tightens, HIBOR rises accordingly and the HKMA cannot offset the impact; the economy absorbs the external rate cycle in full. This constraint was starkly illustrated in 1997–2000: as Hong Kong fell into recession following the Asian financial crisis, it was forced to absorb the Federal Reserve's rate hikes, amplifying the downturn. This paper asks: through which domestic channels does the imported monetary impulse reach output, and with what speed and persistence?

Mainland China: Real shocks instead of Monetary

Following Reform & Opening up by Deng since 1978 Dec, China became the world's manufacturing hub and largest goods trader. As China expands, Hong Kong's role as a gateway for Mainland trade and financial flows deepened correspondingly. Fluctuations in Chinese real GDP therefore transmit directly into Hong Kong's export demand and financial intermediary activity, making China real GDP an important source of external shock to the Hong Kong economy.

Potential Channels

Three domestic channels connect the imported monetary impulse and the China demand shock to Hong Kong real GDP.

The *property sub-channel*: HIBOR increases raise mortgage costs, compress residential property valuations, and tighten collateral constraints (Kiyotaki and Moore, 1997; Iacoviello, 2005; Gerlach and Peng, 2005). In Hong Kong, residential property is the dominant household asset. Declining property prices weigh on GDP growth through the wealth effect and tighter collateral constraints.

The *direct credit sub-channel*: higher HIBOR raises business borrowing costs through the bank lending rate (Bernanke and Gertler, 1995), compressing business investment and private consumption.

The *trade channel*: Hong Kong's deep integration with mainland China means that China's real GDP shapes Hong Kong's domestic economic dynamics. By the mid-2010s, China accounted for over half of Hong Kong's total merchandise export value. This paper captures this channel through exports to China in year-on-year growth terms as an endogenous variable in the BVARX.

3 Data

The dataset covers 113 quarterly observations from 1998 Q1 to 2026 Q1. Due to the 1997 handover and Asian financial crisis, which constitute a structural break in Hong Kong economic development, the start date is selected to be 1998 Q1. Table 1 summarises the eight variables.

The six endogenous variables are: three-month HIBOR (`hibor_3m`), exports to China in year-on-year growth (`hk_exports_china_yoy`), the RVD All Classes property price index in quarter-on-quarter change (`hk_property_price_qoq`), real chain-linked GDP growth (`gdp_growth`), Composite CPI inflation (`cpi_inflation`), and the seasonally adjusted unemployment rate (`unemployment`).

Table 2 reports ADF and KPSS stationarity tests for all model variables. Property prices enter as quarter-on-quarter changes: the level index is $I(1)$ (ADF $p = 0.822$) while

Table 1: Data and Transformations

Variable	Form	Units	Source
<i>Exogenous</i>			
us_ffr	Level	% p.a.	FRED FEDFUNDS, monthly avg. → quarterly
china_gdp	YoY %	%	OECD QNA CHN.B1_GE.GPSA.Q
<i>Endogenous</i>			
hibor_3m	Level	% p.a.	HKMA HIBOR fixing, monthly avg. → quarterly
hk_exports_china_yoy	YoY %	%	C&SD table 410-50013, nominal HKD
hk_property_price_qoq	QoQ %	%	RVD All Classes index, quarterly differenced
gdp_growth	YoY %	%	C&SD table 310-30001, real chain-linked
cpi_inflation	YoY %	%	C&SD table 510-60001, Composite CPI
unemployment	Level	%	C&SD table 210-06101, s.a. M3M

the QoQ transform is stationary (ADF $p < 0.001$). HIBOR in levels is ambiguous (ADF $p = 0.037$, KPSS $p = 0.050$). Since a borderline KPSS result is known to over-reject stationarity in finite samples (Kwiatkowski et al., 1992), and given the economic intuition that HIBOR mean-reverts under the LERS peg, HIBOR is treated as stationary and is therefore excluded from the following Johansen test. Unemployment enters in levels and is I(1)-like; inference is conducted under the levels VAR approach of (Sims et al., 1990), which permits I(1) variables in a VAR without differencing when cointegration is absent.

Table 2: Stationarity Tests

Variable	ADF p	KPSS p	Classification
hk_exports_china_yoy	0.000	0.100	I(0)
gdp_growth	0.016	0.100	I(0)
hk_property_price_qoq	0.000	0.083	I(0)
hibor_3m	0.037	0.050	ambiguous
cpi_inflation	0.153	0.020	I(1)
unemployment	0.082	0.010	I(1)
hk_property_price_idx	0.822	0.010	I(1)

Notes: ADF null: unit root; KPSS null: stationarity.

Table 3 reports Johansen trace and max-eigenvalue tests on the I(1) block. All statistics fall below the 95% critical value, confirming rank zero. These results suggest that VECM/BVECM is not necessary and that BVARX is more suitable for this paper.

Three data limitations are noted. First, exports are measured in nominal Hong Kong dollars; volume and price effects cannot be separated. No suitable price deflator for exports is available. Second, `us_ffr` is a monetary-conditions proxy rather than a clean monetary policy surprise. Third, the post-2010 China-Hong Kong financial integration channel is not directly modelled as no suitable quarterly series extends back to 1998.

Table 3: Johansen Cointegration Tests

H_0 : rank $\leq$	Trace stat	CV 95%	Max-eig stat	CV 95%
0	32.78	35.01	20.22	24.25
1	12.57	18.40	8.89	17.15
2	3.67	3.84	3.67	3.84

Notes: Variables: unemployment, CPI inflation, and property price level index. `det_order=1`, constant outside cointegrating vector.

4 Empirical Methodology

4.1 Models: BVARX

This paper uses a BVARX model with Minnesota prior estimated on quarterly data. The six endogenous variables are: three-month HIBOR, exports to China YoY change, the Hong Kong property price index QoQ change, real Hong Kong GDP growth YoY, CPI inflation YoY, and unemployment rate (level). Following the block-exogenous structure of Cushman and Zha (1997), the US federal funds rate and China’s real GDP growth are included as strictly exogenous variables for the BVARX model. Hong Kong’s economy and monetary base are too small to influence U.S. and Mainland economic conditions, and thus these two variables are assumed to be exogenous.

There are two challenges for classical VAR estimation in this paper: nonstationary data and limited sample size in quarterly data. Also, to retain four lags ($p = 4$) to capture annual changes for quarterly macroeconomic dynamics, classical VAR will overfit with too many coefficients. A BVARX with empirical Bayes Minnesota prior resolves all three: it regularises the overparameterised model by shrinking it toward the prior with ML-optimized means following Giannone et al. (2015), preventing overfit while preserving longer lag structure. BVAR also handles nonstationary data better when the original series are not cleanly classified as $I(1)$ or $I(0)$.

4.2 BVARX Specification

The BVAR models are estimated using Alexandria version 3.0.0 (Legrand, 2026).

$$\mathbf{y}_t = \mathbf{c} + \sum_{\ell=1}^4 \mathbf{A}_\ell \mathbf{y}_{t-\ell} + \mathbf{B}_0 \mathbf{z}_t + \boldsymbol{\varepsilon}_t, \quad \boldsymbol{\varepsilon}_t \sim \mathcal{N}(\mathbf{0}, \boldsymbol{\Sigma}), \quad (1)$$

where $\mathbf{y}_t$ is the 6×1 vector of endogenous variables and $\mathbf{z}_t = (\text{us_ffr}_t, \text{china_gdp}_t)'$ is the block-exogenous forcing vector. The model has $k = 27$ parameters per equation and $T_{\text{eff}} = 109$ observations after conditioning on four initial lags.

Prior. Based on (Litterman, 1986), the Minnesota prior shrinks own-lag coefficients toward a variable-specific AR(1) and cross-variable lags toward zero, with prior variance decaying as $1/\ell^{\pi_3}$ across lags. Hyperparameters are optimized by marginal likelihood maximisation under the Alexandria packages (Legrand, 2026), following Giannone et al. (2015). The optimized parameters are: overall tightness $\pi_1 = 0.085$, cross-variable weight $\pi_2 = 1.0$, lag decay $\pi_3 = 1.0$. The prior’s slackness parameter for exogenous coefficients is set to $\pi_4 = 100$, imposing little prior shrinkage on the exogenous coefficients. The AR prior means δ_i are also ML-optimal, ranging from $\delta = 0.418$ (property prices) to $\delta = 0.991$ (unemployment).

Table 4: Minnesota Prior Hyperparameters

Parameter	Description	Value
π_1	Overall tightness	0.085
π_2	Cross-variable weight	1.000
π_3	Lag decay	1.000
π_4	Exogenous slackness	100.0
δ_{HIBOR}	HIBOR AR prior mean	0.442
δ_{Exports}	Exports AR prior mean	0.627
δ_{Property}	Property AR prior mean	0.418
δ_{GDP}	GDP AR prior mean	0.545
δ_{CPI}	CPI AR prior mean	0.735
$\delta_{\text{Unemployment}}$	Unemployment AR prior mean	0.991

Notes: Values are rounded, selected by marginal-likelihood optimisation.

Posterior inference. The Alexandria package combines the Minnesota prior with the sample likelihood to obtain the posterior distribution of the VAR coefficients, conditional on a fixed OLS estimate of the residual covariance matrix (Legrand, 2026). The program generates 2,000 posterior coefficient draws and calculates the impulse responses for each draw. The reported pointwise 90% credible intervals are the 5th and 95th percentiles of these simulated responses. No stationarity restriction is imposed on the posterior draws.

Exogenous lag structure. The two external variables enter contemporaneously only ($q = 0$). Under the LERS, changes in the U.S. federal funds rate (FFR) are transmitted rapidly to Hong Kong money-market rates. Under quarterly data, extra lags are not regarded as necessary, and delayed domestic adjustment is captured by four lags of the endogenous variables. Adding one lag of the FFR does not eliminate the Ljung–Box rejections in the GDP and CPI equations and leaves most headline responses broadly unchanged. Moreover, China’s real GDP enters as year-on-year growth. Adjacent observations therefore already contain overlapping information. As China’s real GDP is regarded as exogenous, the contemporaneous-only specification is the more appropriate baseline.

4.3 Recursive Identification

The ordering is:

$$HIBOR \rightarrow exports \rightarrow property \rightarrow GDP \rightarrow CPI \rightarrow unemployment \quad (2)$$

Structural shocks are identified via Cholesky decomposition of the reduced-form covariance matrix $\Sigma = \mathbf{P}\mathbf{P}'$, where $\mathbf{P}$ is lower triangular. Variables are assumed not to respond contemporaneously to structural shocks from variables on its right hand side of the (2). All contemporaneous feedback runs downward through the ordering.

The order is arranged based on economic theory and intuition. HIBOR is ordered first: as discussed above under LERS, it adjusts mechanically to the Federal Reserve's target within the quarter. Thus, HIBOR is treated as predetermined. It does not respond to domestic macroeconomic innovations within the same quarter. Exports are ordered second. Hong Kong's exports to China are determined by Mainland import demand and thus its real GDP within the quarter. Domestic HK conditions have minimal impact on how much China buys from HK within the same quarter.

Property precedes GDP because residential property prices are determined in continuous daily markets and respond to monetary and financial conditions faster than quarterly real GDP. Higher property values raise household net worth and relax collateral constraints, transmitting to consumption and investment through the wealth and collateral effects (Iacoviello, 2005; Case et al., 2005). The reverse channel is slower: in Hong Kong, property prices are primarily driven by external economic sentiment and financial conditions rather than domestic income growth, with real GDP found to be weakly exogenous to property prices (Gerlach and Peng, 2005).¹ CPI is ordered fifth: price-setting decisions exhibit within-quarter stickiness, consistent with the monetary VAR literature (Christiano et al., 1999). Unemployment is ordered last: hiring and separation decisions involve adjustment costs that preclude contemporaneous responses to output or financial shocks at the quarterly frequency (Christiano et al., 1999).

Identification is recursive by construction and thus all IRF and FEVD results are conditional on the maintained ordering. To verify that the domestic transmission timing is not an artifact of the Cholesky matrix, this paper also re-evaluate the channel using direct projections with HAC inference and an identical information set.

¹Reversing the property–GDP ordering reduces property's $h = 8$ GDP FEVD share from 20.7% to 2.3%, confirming that this variance attribution is ordering-dependent. However, the property-to-GDP response remains positive with 90% credible intervals excluding zero at $h = 2$ and $h = 3$ under both orderings, supporting the lagged transmission channel.

4.4 Local Projections

Local projections (Jordà, 2005) are estimated for four channels as a timing robustness check. As demonstrated by Plagborg-Møller and Wolf (2021), local projections and VARs estimate identical structural impulse responses in population; therefore, the local projections will yield similar results to the BVAR provided the latter’s finite-lag structure is not severely misspecified. The regression at horizon h is

$$y_{t+h} = \alpha_h + \beta_h s_t + \gamma'_h \mathbf{x}_t + \varepsilon_{t+h}, \quad (3)$$

where s_t is the shock variable and $\mathbf{x}_t$ includes: (i) four lags of all endogenous variables, (ii) current `us_ffrt` and `china_gdpt`, and (iii) contemporaneous values of endogenous variables ordered before s_t in the Cholesky decomposition. The controls in (iii) impose the exact same recursive restrictions as the BVAR, ensuring $\hat{\beta}_h$ carries an identical structural interpretation at each horizon. Standard errors are Newey-West HAC with bandwidth $\max(h+1, n^{1/3})$, where $n = T - h - p$ is the effective sample size at horizon h . The four channels are: HIBOR $\rightarrow$ property, HIBOR $\rightarrow$ GDP, exports $\rightarrow$ GDP, and property $\rightarrow$ GDP.

5 Results

5.1 External Transmission Channels and LERS Validation

Block-exogenous dynamic multipliers validate both external transmission channels. A 1 pp shock to the US federal funds rate (FFR) raises HIBOR by +0.222 pp at $h = 1$ and initially expands Hong Kong GDP by +0.224 pp. This anomalous $h = 1$ expansion reflects a standard small-open-economy demand spillover: US monetary tightening typically coincides with US economic booms, temporarily boosting Hong Kong’s output before lagged financial tightening contracts GDP by -0.196 pp at $h = 4$. Similarly, a 1 pp shock to China’s GDP growth raises exports by +0.571 pp and GDP by +0.219 pp at $h = 1$. These estimates represent unconditional dynamic multipliers, incorporating all endogenous feedback mechanisms within the domestic system.

HIBOR’s own forecast error variance share reaches 94% at $h = 8$, confirming that domestic shocks account for almost none of the variation in local interest rates and validating its position as the first variable in the Cholesky ordering.²

²The within-equation posterior slope on `us_ffr` in the HIBOR equation is +0.526 [+0.428, +0.624]; the lower exo multiplier reflects dampening through feedback from the other endogenous variables (see Table 9).

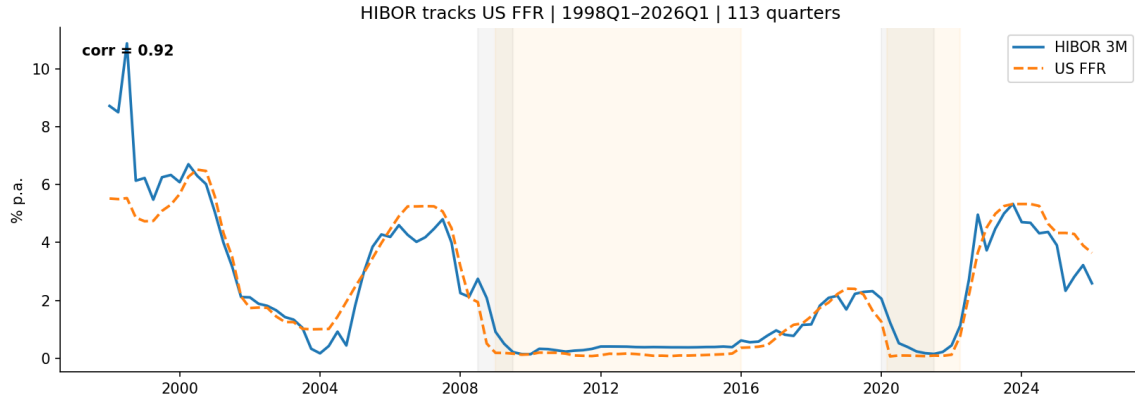


Figure 1: HIBOR and US Federal Funds Rate, 1998Q1–2026Q1

5.2 Dynamic Transmission (Impulse Responses)

Table 5 reports 90% posterior credibility intervals for five channels at $h \in \{1, 2, 4\}$. Figure 2 presents the full 6×6 impulse response matrix.

Table 5: Impulse Responses: 90% Posterior Credibility Bands

Channel	h	Mean	Lower	Upper	Sig.?
HIBOR $\rightarrow$ Property	1	−0.677	−1.004	−0.343	Yes
	2	−0.190	−0.507	+0.106	No
	4	−0.066	−0.208	+0.074	No
Property $\rightarrow$ GDP	1	+0.482	+0.327	+0.637	Yes
	2	+0.406	+0.242	+0.575	Yes
	4	+0.060	−0.066	+0.206	No
HIBOR $\rightarrow$ GDP	1	−0.243	−0.421	−0.069	Yes
	2	−0.309	−0.498	−0.137	Yes
	4	−0.169	−0.293	−0.055	Yes
Exports $\rightarrow$ GDP	1	+0.395	+0.260	+0.528	Yes
	2	+0.210	+0.055	+0.372	Yes
	4	−0.043	−0.198	+0.111	No
GDP $\rightarrow$ CPI	1	+0.124	+0.075	+0.172	Yes
	2	+0.164	+0.102	+0.229	Yes
	4	+0.185	+0.114	+0.259	Yes

Notes: BVARX(4), Minnesota prior, 90% posterior credibility bands. Significance: credibility interval excludes zero.

Monetary transmission and speed asymmetry

A 1 pp HIBOR shock contracts property prices significantly at $h = 1$, with a posterior mean of −0.677 and 90% band $[-1.004, -0.343]$, but loses significance by $h = 2$. Property shocks in turn raise GDP at $h = 1$ (+0.482, $[+0.327, +0.637]$) and $h = 2$ (+0.406,

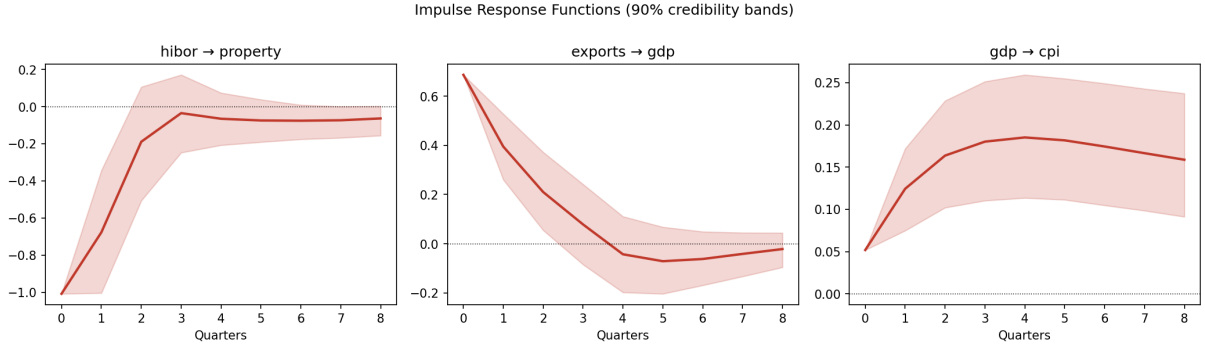


Figure 2: Canonical BVARX(4) Impulse Response Functions, 90% Credibility Bands

$[+0.242, +0.575]$), fading by $h = 4$. The total HIBOR–GDP IRF persists further: -0.243 at $h = 1$, -0.309 at $h = 2$, and -0.169 at $h = 4$, all significant at the 90% level.

This contrast constitutes a speed asymmetry. Floating-rate mortgages link HIBOR directly to financing costs, compressing property prices within the quarter. The resulting wealth effect propagates to GDP at $h = 1$ – 2 before dissipating. The credit and debt-servicing channel accumulates more slowly, sustaining GDP contraction through $h = 4$ (Mishkin, 1996).

The trade channel

Exports predict GDP growth at $h = 1$ ($+0.395$, $[+0.260, +0.528]$) and $h = 2$ ($+0.210$, $[+0.055, +0.372]$), decaying to insignificance at $h = 4$. This channel weakens further after conditioning on current China GDP growth in recursive LP (Table 7), suggesting that Mainland China real activity reaches Hong Kong primarily through contemporaneous block and through trade.

GDP–CPI transmission

GDP predicts CPI significantly from $h = 1$ through $h = 4$, with the posterior mean rising from $+0.124$ to $+0.185$ across horizons ($[+0.114, +0.259]$ at $h = 4$), consistent with demand-pull dynamics in which price-setters respond to realised output conditions with a lag.³

5.3 Forecast Error Variance Decomposition (FEVD)

Table 6 reports the FEVD of GDP growth at $h = 8$. Figure 3 shows the full decomposition for HIBOR, property and real GDP; Figure 4 focuses on the GDP equation. The Cholesky decomposition ordering is: $\text{hibor} \rightarrow \text{exports} \rightarrow \text{property} \rightarrow \text{gdp} \rightarrow \text{cpi} \rightarrow \text{unemp}$.

³The CPI equation carries a confirmed COVID-19 mean break ($p = 0.036$; see Section 6.2). Inflation estimates from 2020–2021 carry additional uncertainty.

Table 6: Forecast Error Variance Decomposition: GDP Growth at $h = 8$

Shock source	Share of GDP variance (%)
Property prices	20.7
Exports to China	16.4
HIBOR shock	8.4
GDP own + other	54.5
Total	100.0

Notes: BVARX(4), Minnesota prior

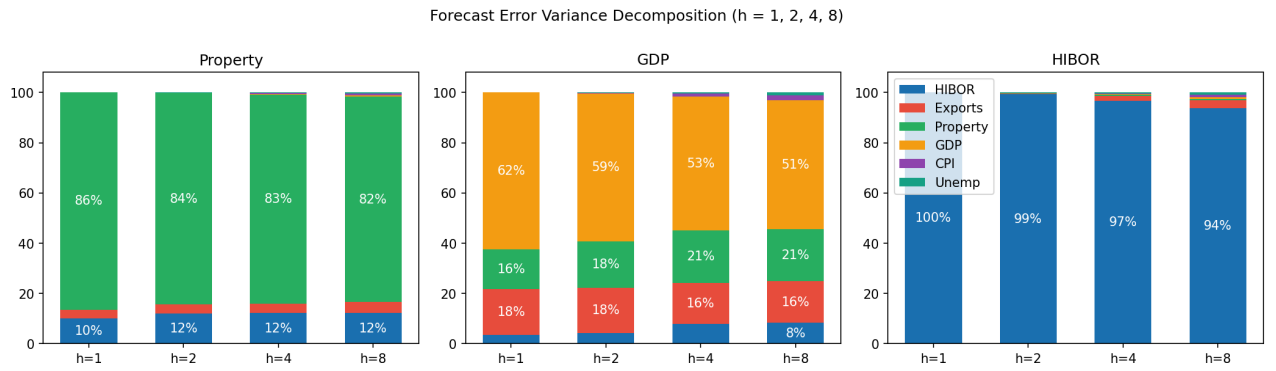


Figure 3: Forecast Error Variance Decomposition, All Equations

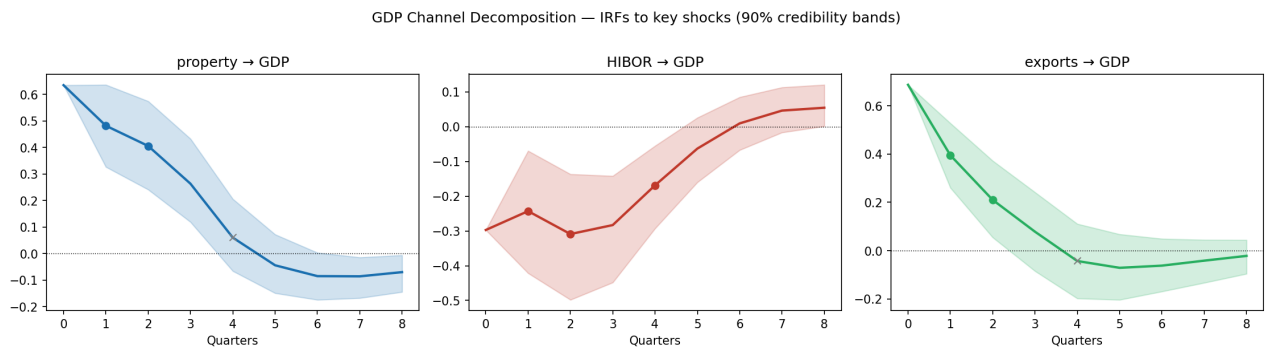


Figure 4: GDP Channel Decomposition: IRFs to Property, HIBOR, and Exports Shocks

Under the current Cholesky decomposition order, under which property prices adjust within the quarter without being affected by contemporaneous shocks in real GDP, property shocks account for the largest share of GDP forecast error variance (20.7%), exceeding the trade channel (16.4%) and the HIBOR shock (8.4%).⁴ HIBOR shocks additionally account for 12% of property forecast error variance at $h = 8$. The full LERS transmission chain is: the US FFR raises HIBOR (94% own-FEVD); HIBOR shocks the property market (12% HIBOR share in property variance); property shocks then propagate to GDP (20.7% of GDP variance). The HIBOR shock contributes 8.4% of GDP forecast error variance; its impulse response remains significant through $h = 4$, sustaining GDP contraction beyond the property amplification window⁵.

6 Robustness and Diagnostics

6.1 Recursive Local Projection

Table 7 and Figure 5 compare local projection (LP) and BVAR significance for four channels. Three of four channels are confirmed across both methods. HIBOR–GDP is confirmed at $h = 1$ and $h = 2$; LP does not confirm the BVAR’s $h = 4$ finding. Exports–GDP is positive in both estimators but falls below the 90% significance threshold in LP at all horizons. This is consistent with the finding in Section 5 that Mainland output reaches Hong Kong primarily through the contemporaneous exogenous block rather than the export channel.

6.2 Structural Stability

Table 8 reports mean-break (Welch t -test) and variance-shift (Brown-Forsythe test; Brown and Forsythe, 1974) results on BVARX(4) residuals at four candidate break dates.

SARS 2003 and China 2015 produce no statistically significant break. The GFC generates variance shifts in HIBOR ($p = 0.036$) and exports ($p = 0.003$) but no GDP mean break ($p = 0.144$). COVID-19 produces a mean break in CPI ($p = 0.036$), a borderline mean break in exports ($p = 0.077$), variance shifts in exports ($p = 0.027$) and unemployment ($p = 0.002$), and no mean break in GDP ($p = 0.260$). The GDP mean break visible under VARX(1) at the GFC ($p = 0.006$) disappears under BVARX(4), consistent with under-specified lag length producing spurious break evidence and Minnesota shrinkage reducing estimation uncertainty (Giannone et al., 2015).

⁴Swapping property and GDP in the Cholesky ordering reduces the property share to 2.3%; the joint (property & GDP own) block is 71.9% under both orderings. The lag-dynamic IRF at $h = 2$ – $h = 3$ is significant in both orderings (Section 6).

⁵The 8.4% is the total orthogonalized HIBOR shock contribution, including both the direct credit route AND the contemporaneous HIBOR $\rightarrow$ property $\rightarrow$ GDP path.

Table 7: Local Projection: Channel Significance at Key Horizons

Channel	h	LP significant (90% HAC)?	BVAR significant (90% CI)?
HIBOR $\rightarrow$ Property	1	Yes	Yes
	2	No	No
HIBOR $\rightarrow$ GDP	1	Yes	Yes
	2	Yes	Yes
	4	No	Yes
Exports $\rightarrow$ GDP	1	No	Yes
	2	No	Yes
Property $\rightarrow$ GDP	1	Yes	Yes
	2	Yes	Yes

Notes: Local projections use the same BVARX information set with contemporaneous recursive controls. Significance: 90% level.

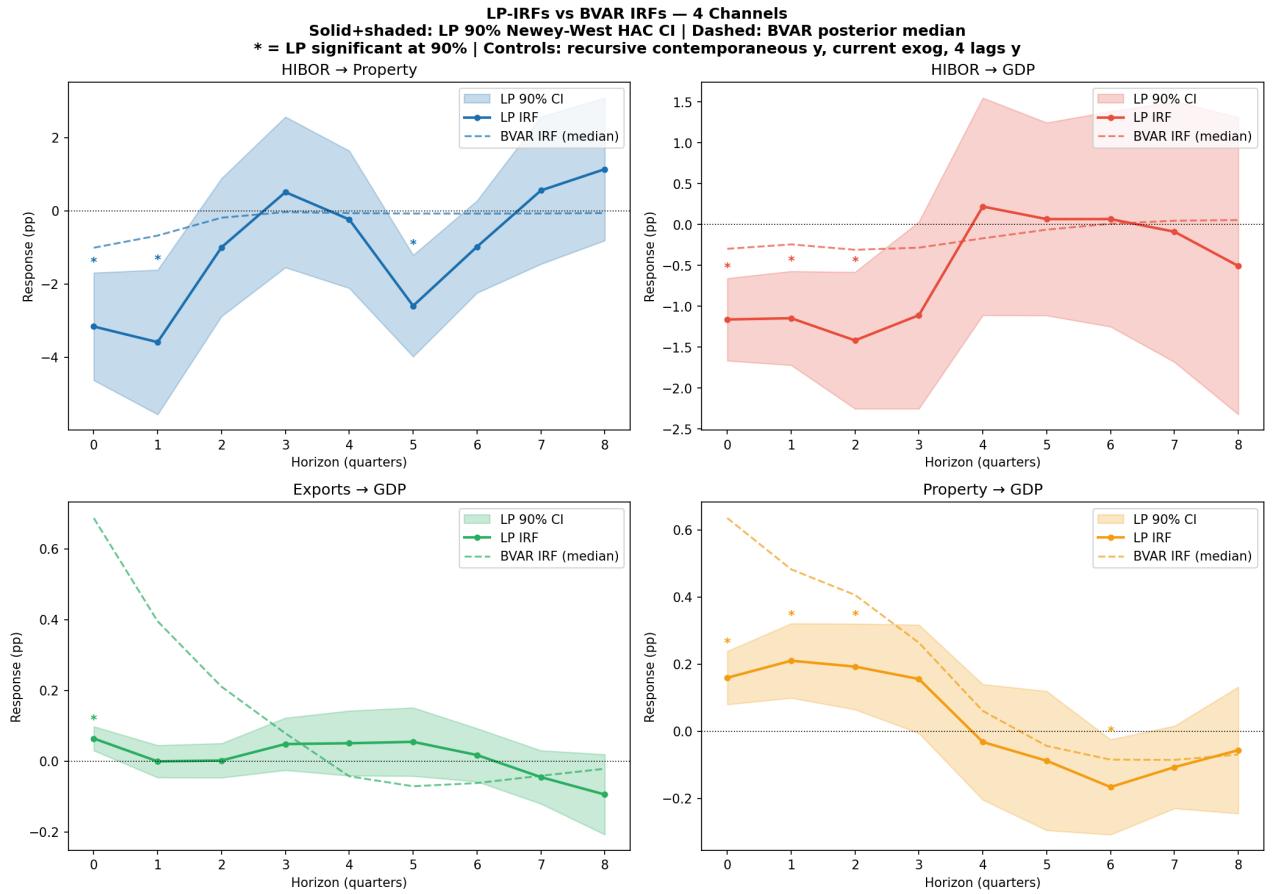


Figure 5: Local Projection vs. BVAR: Impulse Responses for Four Channels

Table 8: Structural Stability: Welch t (Mean) and Brown-Forsythe (Brown and Forsythe, 1974) (Variance) Tests on BVAR(4) Residuals

Equation	SARS 2003	GFC 2008	China 2015	COVID 2020
hibor	stable	var. shift ($p = 0.036$)	stable	stable
exports	stable	var. shift ($p = 0.003$)	stable	borderline ($p = 0.077$); var. shift ($p = 0.027$)
property	stable	stable	stable	stable
gdp	stable	stable ($p = 0.144$)	stable	stable ($p = 0.260$)
cpi	stable	stable	stable	mean break ($p = 0.036$)
unemployment	stable	stable	stable	var. shift ($p = 0.002$)

Bold = mean break; otherwise variance shift or stable ($p > 0.10$).

The CPI COVID mean break ($p = 0.036$) is one confirmed instability; all other rejections are variance shifts, not shifts in the transmission coefficients. Stability in the GDP and property equations supports interpreting the full-sample IRFs as a single transmission regime rather than an average across structurally distinct sub-periods.

6.3 Specification Checks

Unemployment in levels. Re-estimating with first-differenced unemployment leaves the HIBOR–property and property–GDP channels qualitatively unchanged at $h = 2$, with band shifts below 0.1 percentage point and no change in significance. Unemployment is estimated in levels following Sims et al. (1990).

Exogenous lag. Adding one lag of U.S. FFR does not resolve the Ljung-Box failures in the GDP and CPI equations ($p = 0.0001$ and $p = 0.0019$ under $q = 1$, versus $p = 0.0000$ and $p = 0.0023$ under $q = 0$). HIBOR–GDP additionally loses significance at $h = 1$ and $h = 4$ under $q = 1$; all other channels are unchanged. Therefore, the contemporaneous-only specification ($q = 0$) is retained.

6.4 Cholesky Ordering Robustness

Swapping property and GDP in the Cholesky ordering reduces property’s share of GDP forecast error variance from 20.6% to 2.3%; the joint (property & GDP own) block is conserved at 71.9% under both orderings. The drop occurs because Cholesky assigns all within-quarter covariance to whichever variable comes first: under the swap, that co-movement is attributed to GDP, not property.

The IRF at $h = 2$ and $h = 3$ is significant under both orderings (swap point estimates: 0.19 and 0.17). These responses are driven by the lagged A coefficients, which are estimated jointly and are independent of the contemporaneous ordering. Under the swap, the contemporaneous response is zero by Cholesky construction and $h = 1$ (0.129) is insignificant; the 20.6% FEVD reflects this contemporaneous attribution, not the lag dynamics. The contemporaneous ordering is an identification assumption imposed by the researcher; quarterly data cannot identify which variable moves first within the quarter, so the 20.6% figure cannot be empirically defended on that dimension.

6.5 Out-of-Sample Forecast Performance

The BVARX(4) is evaluated against a random walk (RW) and an AR(1) using recursive out-of-sample RMSE ratios (BVARX/benchmark; below 1.0 indicates BVARX superiority) over 28 windows. Against the RW, the BVARX wins on HIBOR (0.831), GDP (0.769), exports (0.976), and unemployment (0.900) at $h = 1$; CPI loses at all horizons (1.108–1.272) and property loses at $h = 4$ (1.177). Against the AR(1), the BVARX wins on HIBOR (0.838), GDP (0.804), and unemployment (0.911) at $h = 1$; exports and property are marginal or negative. All forecasts use realized `us_ffr` and `china_gdp` as exogenous inputs rather than forecasted values, as the exogenous variables are assumed to be unaffected by the endogenous variables in the data-generating process.

For the purpose of fully modelling macroeconomic dynamics in Hong Kong, it would be better to fine-tune the model to improve forecast performance across all variables. However, this particular model provides adequate insights, particularly in forecasting and modelling Hong Kong real GDP.

7 Limitations and Extensions

Recursive identification is order-dependent.

All IRF and FEVD results are conditional on the Cholesky ordering HIBOR, exports, property, GDP, CPI, unemployment. The LERS mechanism provides unusually strong institutional support for placing HIBOR first. The remaining ordering decisions are motivated by economic theory but not uniquely determined; as Section 6.4 shows, the headline 20.6% property FEVD cannot be defended against an alternative contemporaneous structure with statistical theory.

The federal funds rate is not a structural monetary policy surprise.

The model uses the quarterly average federal funds rate as the external monetary-conditions variable. This series conflates anticipated and unanticipated policy changes (Kuttner,

2001) and does not isolate exogenous policy surprises. Results are reduced-form responses to external financial conditions under LERS, not causal effects of identified US monetary policy shocks.

Block exogeneity is tested; federal funds rate result is inconclusive.

The model follows Cushman and Zha (1997) in treating `us_ffr` and `china_gdp` as strictly exogenous. No feedback from Hong Kong variables to the external block is permitted. A Granger-causality test (Wald, VAR(4)) fails to reject block exogeneity for China GDP ($p = 0.250$); the null for the federal funds rate is marginally rejected ($p = 0.046$), but chi-sq inference is invalid with I(1) variables.

The assumption is based on economic theory: Hong Kong cannot plausibly cause movements in US monetary policy or China real GDP at quarterly frequency.

Forecast error variance decomposition does not capture contributions of external variables.

The FEVD reported in Table 6 covers only the endogenous recursive shock decomposition. The contributions of U.S. FFR and China real GDP to Hong Kong GDP variance, the external channels this paper set out to quantify, are excluded from the FEVD; their variance is conditioned out of the endogenous residuals before Cholesky factorization. Section 5 reports dynamic multipliers for the U.S. FFR and China GDP growth.

China–Hong Kong financial integration is only partially captured.

The model captures Mainland spillovers through China GDP growth and Hong Kong exports to China, but it does not directly observe China-related financial intermediation channels such as offshore listings, IPO activity, or cross-border capital-market issuance. Those channels became more important to Hong Kong after 2010, but no reliable full-sample quarterly series was available for inclusion in the model. As a result, the estimated China channel should be read primarily as a real-demand and trade channel. This likely causes the paper to understate the full effect of Mainland shocks on Hong Kong.

GDP and CPI residual autocorrelation remains.

The GDP and CPI equations fail the Ljung-Box white-noise test; adding a lag of the federal funds rate does not resolve this. A fixed-coefficient model produces runs of same-signed forecast errors that Ljung-Box detects as autocorrelation under a structural break. GFC and COVID-19 each produced multi-quarter directional runs in output and inflation that a fixed-coefficient model cannot fully track (Section 6.2). A time-varying parameter VAR (Primiceri, 2005) would better accommodate these regime shifts, but is data-demanding relative to the 113-quarter sample.

Sample spans major structural breaks.

The 28-year sample covers the Asian financial crisis aftermath, the GFC, the 2019 social unrest, and COVID-19. The CPI equation has a confirmed COVID-19 mean break ($p = 0.036$); the fixed-coefficient specification cannot accommodate discrete regime changes. Inflation transmission results from 2020–2021 are thus unreliable. IRF and FEVD estimates represent average dynamics over the full sample and may not hold under further regime changes.

7.1 Extension: Zero Lower Bound Asymmetry

Because Hong Kong’s peg mechanically transmits US policy rates to HIBOR (Section 4.3), and because the HIBOR–property channel is the fastest domestic transmission link in our estimates (Section 5), the zero lower bound presents a natural test of whether that channel weakens when the federal funds rate is near zero. A state-dependent LP-IRF splits the sample into normal-rate and near-zero-rate (ZIRP) regimes. Let $\text{ZIRP}_t = 1$ if the federal funds rate is below 0.25%, covering 36 of 113 observations from 2009 Q1 to 2022 Q1. At $h = 1$:

$$\begin{aligned}\hat{\beta}_{\text{normal}} &= -2.686 \quad 90\% \text{ CI: } (-3.796, -1.576) \quad [\text{significant}], \\ \hat{\beta}_{\text{ZIRP}} &= -0.514 \quad 90\% \text{ CI: } (-7.913, +6.884) \quad [\text{not significant}].\end{aligned}$$

The normal-regime estimate is precise and significant. The ZIRP point estimate is smaller in magnitude, consistent with weaker transmission, but the 90% CI is too wide to support any conclusion. During ZIRP periods HIBOR is pinned near zero and barely moves, so the data contain too little variation in HIBOR to identify significantly how property prices respond, producing the wide confidence interval.

8 Conclusion

Under the Linked Exchange Rate System, Hong Kong’s three-month interbank rate follows the federal funds rate within the quarter. A rate rise reduces property prices immediately — 90% posterior band $[-1.004, -0.343]$ at $h = 1$ — but the property response crosses zero by $h = 2$ and is negligible thereafter. The direct route to output is slower: the posterior band excludes zero at $h = 1$, $h = 2$, and $h = 4$, consistent with a credit and mortgage drag that persists after property prices have stabilised. The two sub-channels thus operate at different speeds: the property route amplifies within a quarter but dissipates by $h = 2$; the credit and mortgage channel sustains the contraction through a full year. Property is the dominant domestic amplifier of the US monetary shock, accounting for 20.7% of GDP forecast-error variance at $h = 8$ against 8.4% for the direct HIBOR shock; the 20.7% is

ordering-conditional, falling to 2.3% under an alternative property–GDP ordering, though the impulse response remains significant at $h = 2$ and $h = 3$ under both.

A one-percentage-point rise in China GDP growth raises Hong Kong output by 0.22 percentage points at $h = 1$, fading by $h = 4$. The domestic amplifier for this external demand shock is exports: the endogenous export-to-GDP impulse response is significant at $h = 1$ and $h = 2$, accounting for 16.4% of GDP forecast-error variance at $h = 8$. The model produces lower out-of-sample forecast errors for real GDP at one to four quarters ahead relative to a AR and RW benchmark.

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A Posterior Coefficient Matrix

Table 9 reports posterior means for all 27×6 coefficients of the BVARX(4) system (seed = 67, 2,000 draws, 90% credibility level). Rows are ordered: constant, two exogenous variables (*us_ffr*, *china_gdp*), then four lags of six endogenous variables. An asterisk (*) denotes that the 90% posterior credibility interval excludes zero.

Table 9: BVARX(4) posterior coefficient means. Equation ordering (columns): HIBOR $\rightarrow$ Exports $\rightarrow$ Property $\rightarrow$ GDP $\rightarrow$ CPI $\rightarrow$ Unemployment (Cholesky identification ordering). * = 90% posterior credibility interval excludes zero. Minnesota prior: $\pi_1 = 0.085$, $\pi_2 = \pi_3 = 1.0$. AR decay coefficients: $[0.442, 0.627, 0.418, 0.545, 0.735, 0.991]$.

	HIBOR	Exports	Property	GDP	CPI	Unemp
Constant	-0.008	-0.553	-1.136	-2.227*	+1.397*	+0.419*
<i>us_ffr</i>	+0.526*	+0.599	-0.023	+0.536*	+0.053	-0.135*
<i>china_gdp</i>	-0.023	+0.837*	+0.301*	+0.395*	+0.089*	-0.010
HIBOR _{<i>t</i>-1}	+0.440*	-0.356	-0.553	-0.068	-0.084	+0.050
Exports _{<i>t</i>-1}	-0.005	+0.630*	+0.005	-0.001	+0.001	+0.003
Property _{<i>t</i>-1}	+0.003	+0.040	+0.421*	+0.051	+0.007	-0.010*
GDP _{<i>t</i>-1}	-0.002	+0.106	-0.057	+0.545*	+0.031	-0.027*
CPI _{<i>t</i>-1}	-0.008	-0.147	-0.069	-0.119	+0.736*	-0.023
Unemp _{<i>t</i>-1}	+0.025	+0.171	-0.072	+0.182	-0.422*	+0.990*
HIBOR _{<i>t</i>-2}	+0.000	-0.313	+0.283	-0.244*	-0.033	+0.041*
Exports _{<i>t</i>-2}	-0.002	+0.019	+0.004	-0.006	+0.004	-0.001
Property _{<i>t</i>-2}	+0.001	+0.028	-0.063*	+0.025	+0.007	+0.000
GDP _{<i>t</i>-2}	+0.001	-0.062	-0.018	+0.032	-0.003	+0.005
CPI _{<i>t</i>-2}	+0.014	-0.435	+0.091	-0.048	-0.000	+0.009
Unemp _{<i>t</i>-2}	+0.020	+0.324	+0.101	-0.223	+0.041	-0.055
HIBOR _{<i>t</i>-3}	-0.014	-0.113	-0.030	-0.088	-0.039	+0.018
Exports _{<i>t</i>-3}	-0.001	-0.022	+0.001	-0.003	+0.001	-0.000
Property _{<i>t</i>-3}	+0.002	+0.015	-0.009	+0.013	+0.003	+0.002
GDP _{<i>t</i>-3}	-0.002	-0.071	+0.013	-0.030	-0.004	+0.002
CPI _{<i>t</i>-3}	+0.008	-0.059	+0.100	-0.004	+0.002	+0.001
Unemp _{<i>t</i>-3}	+0.020	-0.361	-0.080	+0.173	+0.021	-0.005
HIBOR _{<i>t</i>-4}	-0.001	+0.141	-0.079	-0.017	-0.041	+0.018
Exports _{<i>t</i>-4}	-0.000	-0.034*	+0.006	-0.004	+0.002	+0.000
Property _{<i>t</i>-4}	+0.000	-0.019	+0.011	-0.010	+0.000	+0.001
GDP _{<i>t</i>-4}	-0.000	-0.099	+0.039	-0.045*	+0.001	+0.003
CPI _{<i>t</i>-4}	+0.007	-0.065	+0.021	+0.044	-0.009	-0.001
Unemp _{<i>t</i>-4}	+0.010	-0.378	+0.064	-0.009	+0.011	+0.005